

**BRIEF COMMUNICATION**

COVID-19 Vaccine-Related Movement Disorders: A Systematic Review

Grace Elysse D. Angeles,¹ Lowrence Precious C. Dichoso,¹ Roland Dominic G. Jamora^{1,2} ✉

¹Division of Adult Neurology, Department of Neurosciences, College of Medicine and Philippine General Hospital, University of the Philippines Manila, Manila, Philippines

²Section of Neurology, Department of Internal Medicine, Cardinal Santos Medical Center, San Juan City, Philippines

ABSTRACT

Objective Since the release of vaccines against coronavirus disease 2019 (COVID-19), there have been reports of vaccine-related neurologic complications. This study aimed to perform a descriptive systematic review of movement disorders associated with COVID-19 vaccines.

Methods We described the demographics, clinical presentation, management, outcomes, and proposed pathomechanism of the patients. A systematic review was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. A standardized tool was used to assess the quality of the cases.

Results We identified 8 articles that met our inclusion criteria; these articles included 10 patients who developed movement disorders after vaccination. The majority were males ($n = 8$), with a median age of 64.5 years. The most common movement disorder was hemichorea. The rest presented with generalized chorea with myoclonus, cervical dystonia, and akathisia. Most patients respond to immunotherapy. The standardized tool used showed that most studies had a low risk of bias.

Conclusion The reported incidence of vaccine-related movement disorders was low based on available published cases.

Keywords COVID-19; Vaccination; Movement disorders.

INTRODUCTION

Severe acute respiratory syndrome coronavirus-2, the strain that causes coronavirus disease 2019 (COVID-19), primarily affects the respiratory system and led to the declaration of a global pandemic in March 2020.¹ The development of vaccines against this novel virus was expedited to prevent further spread of the disease. As of August 2023, 12 vaccines have been granted emergency use listing by the World Health Organization (WHO), and more than 13.1 billion vaccines have been administered worldwide (<https://extranet.who.int/prequal/vaccines/covid-19-vaccines-who-emergency-use-listing>).² The documentation of rare adverse events makes it important for risk-benefit

evaluations due to the lack of available clinical trials.³ Notably, neurological complications following COVID-19, such as movement disorders, have been reported in case reports with a frequency of 0.00002–0.0002.⁴

Several studies have reported movement disorders associated with ongoing or previous COVID-19 infection. However, few studies have investigated COVID-19 vaccine-related movement disorders. Despite the large databases that were established to detect adverse effects of the COVID-19 vaccine, detailed documentation on the identified movement disorders was not included.

While case reports may be limited by their ability to establish generalizations, they provide a more comprehensive investigation to confirm that movement disorders are more likely to result

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✉ Corresponding author: Roland Dominic G. Jamora, MD, PhD

Division of Adult Neurology, Department of Neurosciences, College of Medicine and Philippine General Hospital, University of the Philippines Manila, Zone 72, 670 Padre Faura St, Ermita, Manila 1000, Philippines / Tel: +62-385548400 / E-mail: rgjamora@up.edu.ph

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from COVID-19 vaccination. A systematic review of case reports may have a significant role in documenting the clinical presentation, ancillary tests performed, neuroimaging, management and outcome to establish that the movement disorder was indeed vaccine related. This review aimed to provide thorough insight into the literature on movement disorders following COVID-19 vaccination.

MATERIALS & METHODS

This systematic review was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The study was registered in the PROSPERO database of systematic reviews (PROSPERO registration number: CRD42023463797).

The researchers considered case series and case reports that included patients who developed movement disorders following any COVID-19 vaccine. We included all possible movement disorders, including hyperkinetic and hypokinetic movement disorders.

We included only case reports with detailed descriptions of the neurologic symptoms, vaccine received, symptom onset, cranial imaging, management, outcome and possible mechanism of vaccine-related adverse events. There were no restrictions in terms of age, sex or ethnicity of the population. We excluded studies that included patients with prior or ongoing COVID-19 infection, preexisting movement disorders, a history of neurodegenerative disease, functional neurologic disorder or encephalitis who developed abnormal movements during the course of illness.

We systematically searched for case reports and case series in the PubMed, EBSCO, Scopus, and WHO COVID-19 Research databases. We used the following general and MeSH terms: ["Movement disorders" OR "hyperkinetic" OR "hypokinetic" OR "dyskinesia" OR "myoclonus" OR "chorea" OR tremor" OR "tics" OR "ballism" OR "athetosis" OR "Parkinson's disease" OR "Parkinsonism" OR "akathisia" OR "restless leg syndrome" OR "stiff person syndrome"] AND ["COVID-19 vaccine" OR "COMIRNATY BioNTech" OR "VAXZEVRIA AstraZeneca" OR "COVISHIELD ChAdOx1-S" OR "SPIKEVAX Moderna Biotech" OR "Inactivated COVID-19 Vaccine" OR "CoronaVac Sinovac" OR "COVAXIN" OR "COVOVAX" OR "NUVAXOVID" OR "CONVIDECIA" OR "SKYCovione (GBP510)"]. The literature search was limited to studies published from the inception of records in these major databases until July 2023.

Studies were identified through database searching and manual searching, and those that did not meet the inclusion criteria were excluded. The following data were extracted: patient de-

mographics, descriptions of the movement disorder, vaccine received, symptom onset, cranial imaging, management, outcome and possible mechanism of vaccine-related adverse events.

Case reports and case series have a high likelihood of bias. To address this issue, the researchers adapted a standardized tool to assess quality according to four domains: selection, ascertainment, causality, and reporting (Supplementary Table 1 in the online-only Data Supplement).⁵ The sum of the scores is aggregated and classified as low, moderate, and high risk.

RESULTS

A PRISMA flow diagram of the selected studies is presented in Figure 1. A total of 363 studies were identified, 32 duplicates were removed, and 301 studies were excluded.

Eight articles contained 10 individual patients who met the inclusion criteria; 8 were male, and the median age was 64.5 years. Five of the patients did not report any comorbid medical conditions. Four individuals received the Pfizer BioNTech COVID-19 vaccine, two received the AZD1222 vaccine, two received the BBIBP-CorV (Sinopharm) vaccine, and the remaining received the ChAdOx1 nCoV-19 vaccine ($n = 1$) and Moderna Biotech ($n = 1$). Three patients experienced movement disorders after receiving their second dose of vaccine, and the remaining patients ($n = 7$) experienced symptoms after receiving their first dose. Symptoms presented as early as 12 hours, and the latest symptoms were reported after 40 days of vaccination. The majority ($n = 7$) reported hemichorea as the predominant movement disorder after receiving the COVID-19 vaccine (Table 1).⁶⁻¹³ The remaining patients reported generalized chorea with myoclonus, cervical dystonia, akathisia, and hemiballismus. Cranial imaging was mostly unremarkable for the cases reported. One of the reported cases involved vaccine-induced thrombocytopenia resulting in an ischemic infarct in the thalamus.⁶ Another case reported the presence of voltage-gated potassium channel and glutamic acid decarboxylase serum antibodies, possibly explaining an autoimmune-mediated movement disorder.⁷

The possibility of an autoimmune post-vaccination reaction was the proposed pathophysiology for vaccine-related movement disorders. Methylprednisolone and intravenous immunoglobulin therapy showed promising results in seven of the cases reported. In one case report, tetrabenazine was used but was eventually discontinued due to the absence of any effect, after which the patient was switched to methylprednisolone therapy.⁸ Other therapeutic options that have been used include clonazepam and botulinum toxin for the treatment of dystonia and hemichorea, haloperidol for hemichorea, and ropinirole for generalized chorea. Another patient reported vaccine-induced

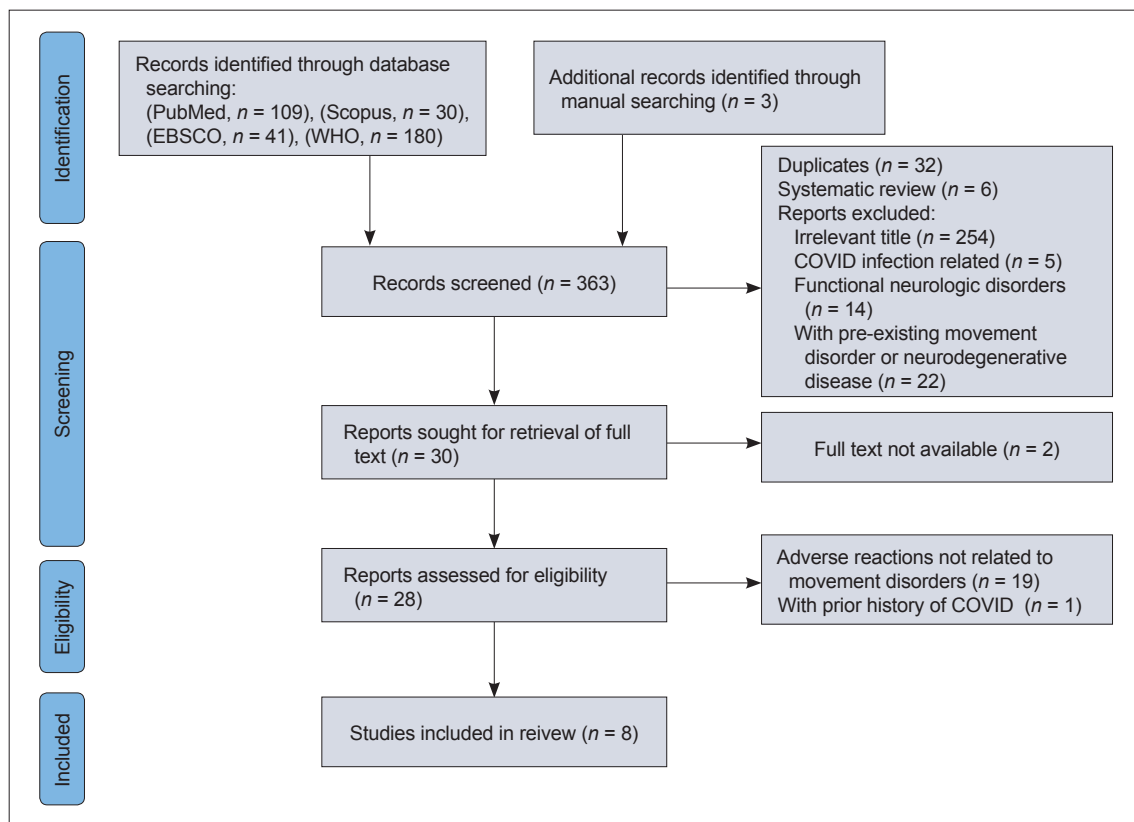


Figure 1. PRISMA flow diagram of included studies. WHO, World Health Organization; COVID, coronavirus disease; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

prothrombotic immune thrombocytopenia based on cranial imaging and hence was given anticoagulants.⁶

The researchers used a standardized tool to assess the methodological quality of the studies (Supplementary Table 1 in the online-only Data Supplement).⁵ One case report was determined to have a high risk of bias and did not include any neuroimaging results or other ancillary tests to rule out other differential diagnoses.

DISCUSSION

Over 13.5 billion vaccines have been administered by 2023; however, there is a lack of documentation on COVID-19 vaccine-related movement disorders.² After a thorough search, this review identified 10 cases of vaccine-related movement disorders. The majority of the patients were males, with a median age of 64.5 years, who received their first dose of the COVID-19 vaccine. In the studies reviewed, hemichorea was the most common movement disorder, and autoimmunity was the proposed pathomechanism.

Hemichorea was the predominant movement disorder described as slow, irregular, and purposeless involuntary move-

ments of the arms and legs.¹³ According to a pharmacovigilance database, approximately 40 cases of hemichorea and hemiballismus have been attributed to COVID-19 vaccination; however, the details of these cases are unknown.⁸ This study also revealed generalized chorea with myoclonus, cervical dystonia, and akathisia after COVID-19 vaccination. These movement disorders have also been reported in pharmacovigilance reports; however, there are no details on the associated workup, treatment or outcome.⁴

Although cranial imaging has a role in the diagnosis of patients with movement disorders, cranial magnetic resonance imaging (MRI) was unremarkable in almost all of the patients who presented. Nonetheless, one patient showed abnormalities in functional imaging, showing perfusion pattern asymmetry on brain single photon emission computed tomography, hence detecting early changes in the uptake of tracers in the early clinical stages of disease.¹³ In another case, a vaccine-induced prothrombotic thrombocytopenia showed ischemia on cranial MRI strategically located on the thalamus.⁶

With the lack of clear identifiable laboratory evidence pointing to a different etiology and the universal response to immunotherapy, an autoimmune cause was the proposed pathomechanism in the identified case reports. An abnormal immunological

Table 1. Clinical features of patients with COVID-19 vaccine related movement disorders

Study	Movement disorder	Age (yr)/sex	Comorbidities	Vaccine	Symptom onset since vaccination	Neuroimaging	Management	Outcome	Proposed Pathophysiology
Shahali et al., ⁶ 2022	Right sided hemichorea	72/M	None	First dose ChAdOx1 nCoV-19 vaccine	9 days	Cranial MRI: left thalamic ischemic infarct	Nadroparin 90 U/kg/day sodium ozagrel 160 mg/day edaravone 60 mg/day methylprednisolone 1 g/kg haloperidol 0.75 mg/kg/day warfarin 0.125 mg/kg/day	Improved after 2 weeks	Vaccine-induced prothrombotic immune thrombocytopenia
Dulanto et al., ⁷ 2023	Generalized chorea with myoclonus	57/F	Type 2 diabetes mellitus, hypertension, dyslipidemia, gastritis, major depressive disorder, generalized anxiety disorder, and chronic substance abuse	Second Moderna booster	14 days	Cranial MRI: unremarkable	Methylprednisolone plasmapheresis rituximab 375 mg/m ² ropinirole (dose not specified)	Chorea and myoclonus improved after immunotherapy	Positive for voltage-gated potassium channel and glutamic acid decarboxylase antibodies pointing toward possible autoimmunity and molecular mimicry
Batot et al., ⁸ 2022	Left hemichorea-hemiballismus	90/M	Hypertension, atrial fibrillation, prior myocardial infarction	Second dose of the Pfizer-BioNTech	Within hours after second dose, 21 days after first dose	Cranial MRI: no abnormal signals in the basal ganglia FDG-PET: increased metabolism of right putamen (compared to the left)	Tetrabenzine 75 mg daily × 3 weeks Oranzapine 7.5 mg daily for 1 week Methylprednisolone 1 g per day for 5 days	Abnormal movements significantly improved 10 days after methylprednisolone with resolution of asymmetric metabolism of the putamen on FDG-PET	Dysimmune reaction induced by COVID-19 vaccination
Algahtani et al., ⁹ 2022	Cervical dystonia with left laterocollis	38/M	None	First dose Pfizer-BioNTech BNT16B2b2	24 hours	Cranial and cervical spine MRI: unremarkable	Oral clonazepam 0.5 mg/day botulinum toxin (dose not specified)	Responded transiently to medical therapy	Autoantibody-mediated immunological dysfunction
Salinas and Dieppa, ¹⁰ 2021	Akathisia	36/F	Atopic dermatitis, allergic rhinitis, anxiety	Second dose Pfizer-BioNTech vaccine	12 hours	None	Spontaneously resolved	Spontaneously resolved	Extrapyramidal system involvement and immune response of triggered by vaccine
Matar et al., ¹¹ 2021	Left hemichorea-hemiballismus	88/M	Dyslipidemia, hypertension, gout	First dose of AZD1222	16 days	Cranial MRI: unremarkable	Intravenous methylprednisolone 1 g for 3 doses	Significant resolution within 24 hrs of first dose of methylprednisolone	Autoimmune postvaccination reaction
Matar et al., ¹¹ 2021	Left hemichorea-hemiballismus	84/M	Asthma, allergic aspergillosis, primary orthostatic hypertension, treated colorectal, esophageal and prostate carcinoma	First dose of AZD1222	40 days	Cranial MRI: unremarkable	Intravenous methylprednisolone 1 g for 3 doses	Partial resolution after first dose of methylprednisolone, but resolved after 3 days of corticosteroids	Autoimmune postvaccination reaction

Table 1. Clinical features of patients with COVID-19 vaccine related movement disorders (continued)

Study	Movement disorder	Age (yr)/sex	Comorbidities	Vaccine	Symptom onset since vaccination	Neuroimaging	Management	Outcome	Proposed Pathophysiology
Salari and Etemadifar, ¹² 2022	Right sided hemichorea	13/M	None	First dose of BBIBP-CorV (Sinopharm) inactivated virus	7 days	Cranial MRI: nonspecific multiple white matter lesions	Intravenous methylprednisolone 1 g/d for 3 days followed by oral prednisolone 50 mg/d and tetrabenazine 25 mg/d	Improved after 1 month follow-up	Post vaccination inflammatory mechanism also seen on as white matter changes on MRI
Salari and Etemadifar, ¹² 2022	Left sided hemichorea	18/M	None	BBIBP-CorV (Sinopharm) inactivated virus	7 days	Cranial MRI: nonspecific white matter lesions	Intravenous methylprednisolone 1 g/d for 3 days followed by oral prednisolone 50 mg/d and tetrabenazine 25 mg/d	Mild choreic movements after 1 month follow-up	Post vaccination inflammatory mechanism
Ryu et al., ¹³ 2022	Right sided hemichorea	83/M	None	Second dose of Pfizer-BioNTech COVID-19 vaccine	1 day	Cranial MRI: unremarkable Brain SPECT: perfusion pattern asymmetrically decreased in left thalamus	Haloperidol 0.75 mg two times a day	Relieved symptoms at two weeks of follow-up	Unclear pathophysiology

COVID-19, coronavirus disease 2019; SPECT, single photon emission tomography; MRI, magnetic resonance imaging; FDG-PET, fluorodeoxyglucose-positron emission tomography.

attack resulting in neuronal dysfunction and manifesting as a movement disorder could be one of the most plausible causes of vaccine-induced movement disorders, similar to adult-onset autoimmune chorea and hemichorea.^{14,15} Moreover, in addition to the COVID-19 vaccine, one patient reported acute chorea following vaccination against human papilloma virus due to molecular mimicry.¹⁶ This autoimmune theory was also proposed for patients who had hyperkinetic disorders after having COVID-19.¹⁷

There were very few patients with vaccine-related movement disorders other than those caused by the COVID-19 vaccine. A literature search revealed one 30-year-old woman with post-vaccine-related opsoclonus-myoclonus syndrome after rubella vaccination. The proposed pathomechanism was due to molecular mimicry resulting from the cross-reactivity of cerebral antigens and viral antigens.¹⁸ In addition, postvaccine parkinsonism was documented after receiving anti-measles and antitetanus vaccines.⁴

In future studies, patients with well-established functional neurologic disorders with movement disorder manifestations and patients with encephalitis as an adverse event of COVID-19 vaccination with a clinical presentation of abnormal movement may be included.

Conclusion

Movement disorders as a complication of COVID-19 vaccination are rare. In this systematic review, hemichorea was identified as the most common movement disorder after vaccination. This descriptive systematic review highlighted the autoimmune pathomechanism involved in the development of movement disorders, and treatment with immunotherapy resolved symptoms.

Supplementary Materials

The online-only Data Supplement is available with this article at <https://doi.org/10.14802/jmd.24001>.

Conflicts of Interest

The authors have no financial conflicts of interest.

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Author Contributions

Conceptualization: Grace Elysse D. Angeles, Roland Dominic G. Jamora. Data curation: all authors. Formal analysis: Grace Elysse D. Angeles, Lowrence Precious C. Dichoso. Investigation: Grace Elysse D. Angeles, Lowrence Precious C. Dichoso. Methodology: Grace Elysse D. Angeles, Roland Dominic G. Jamora. Supervision: Roland Dominic G. Jamora. Validation: all authors. Visualization: Grace Elysse D. Angeles. Writing—original draft: Grace Elysse D. Angeles, Roland Dominic G. Jamora. Writing—review & editing: Grace Elysse D. Angeles, Roland Dominic G. Jamora.

ORCID iDs

Grace Elyse D. Angeles <https://orcid.org/0000-0003-3245-4368>
 Lowrence Precious C. Dichoso <https://orcid.org/0000-0001-7813-4146>
 Roland Dominic G. Jamora <https://orcid.org/0000-0001-5317-7369>

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